Bubble sort:

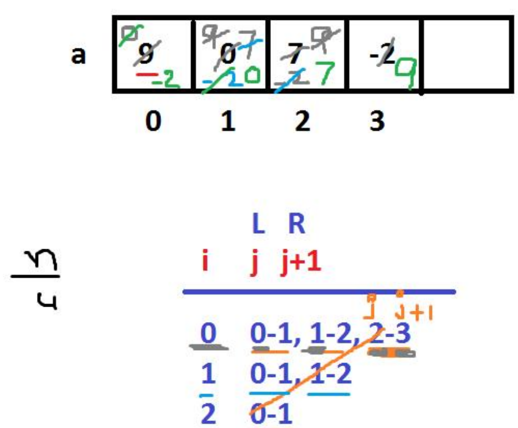
a	9	0	7	-2	
	0	1	2	3	

9

9	0	7	-2
0	9	7	-2
0	7	9	-2
0	7	-2	9
0	-2	7	9
-2	0	7	9

i	j (L)	j+1 (R)
0	0-1, 1-2, 2-3	
1	0-1, 1-2	
2	0-1	

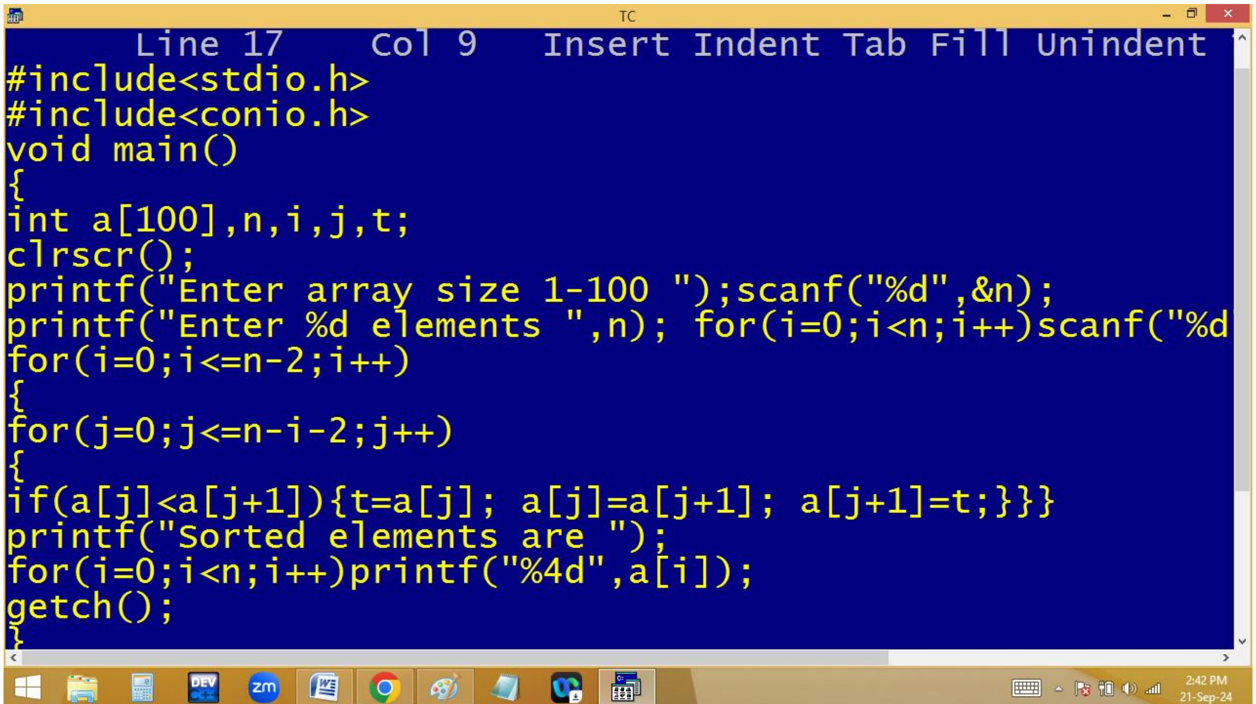
```
for( i=0; i<=n-2; i++)
{
for( j=0; j<=n-i-2; j++)
{
         L        R
 if(a[j]>a[j+1])
 {
 t=a[j];a[j]=a[j+1];a[j+1]=t;
 }
 }
 }
```



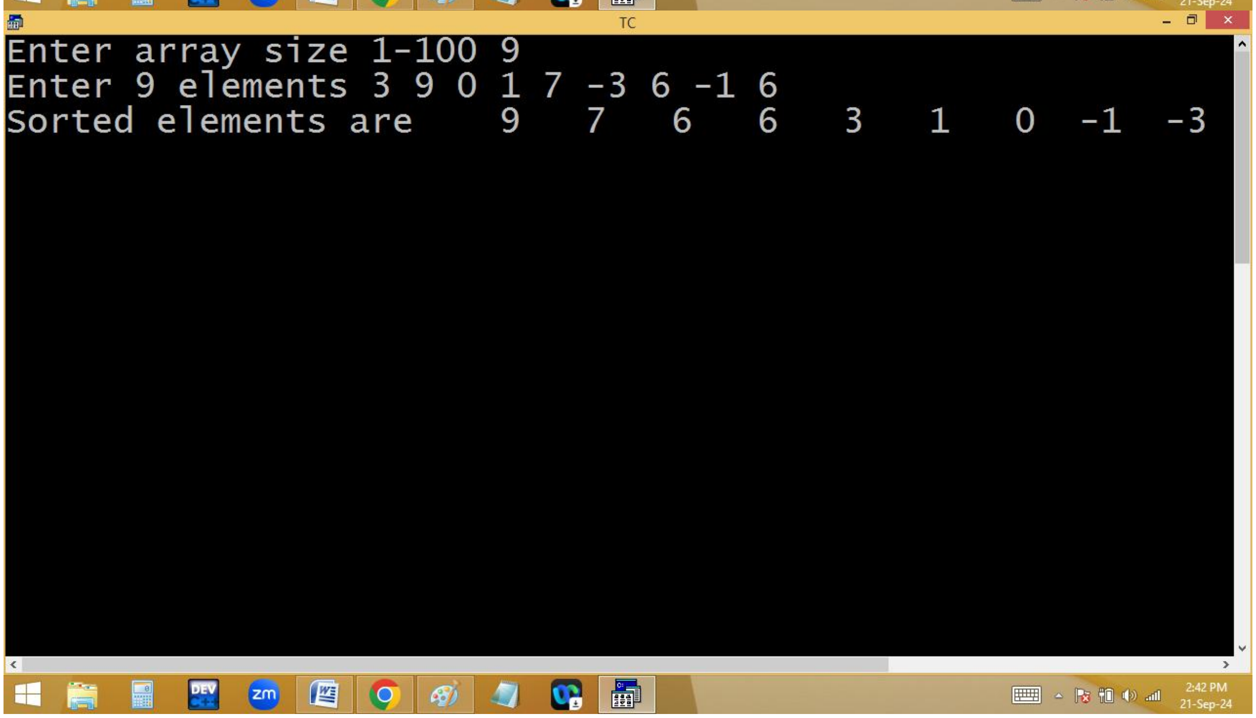
```
        Line 16     Col 1    Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,j,t;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n); for(i=0;i<n;i++)scanf("%d
for(i=0;i<=n-2;i++)
{
for(j=0;j<=n-i-2;j++)
{
if(a[j]>a[j+1]){t=a[j]; a[j]=a[j+1]; a[j+1]=t;}}}
printf("Sorted elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

```
Enter array size 1-100 9
Enter 9 elements 3 9 0 4 -3 7 4 -1 8
Sorted elements are   -3   -1    0    3    4    4    7    8    9
```

Descending order:



```
        Line 17     Col 9    Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,j,t;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n); for(i=0;i<n;i++)scanf("%d
for(i=0;i<=n-2;i++)
{
for(j=0;j<=n-i-2;j++)
{
if(a[j]<a[j+1]){t=a[j]; a[j]=a[j+1]; a[j+1]=t;}}}
printf("Sorted elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

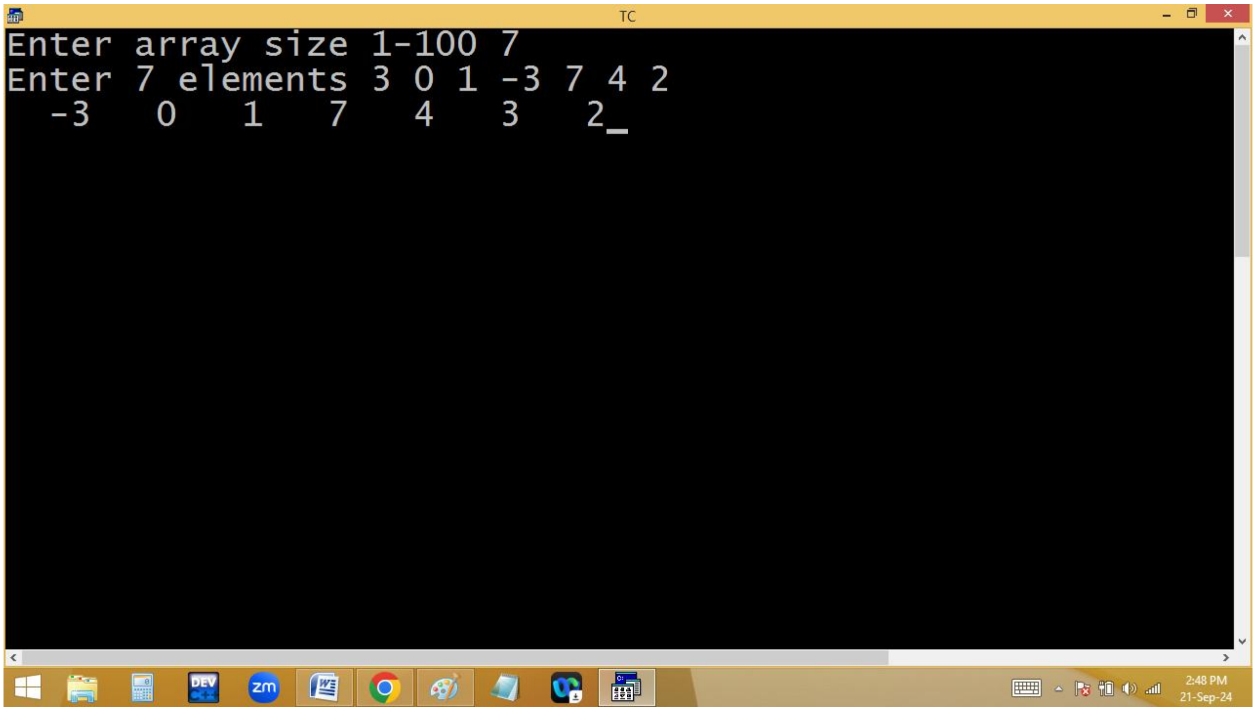


```
Enter array size 1-100 9
Enter 9 elements 3 9 0 1 7 -3 6 -1 6
Sorted elements are    9   7   6   6   3   1   0  -1  -3
```

Half elements in ascending and half descending order:

```
      Line 17     Col 9    Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,j,t;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n); for(i=0;i<n;i++)scanf("%d
for(i=0;i<=n-2;i++)
{
for(j=0;j<=n-i-2;j++)
{
if(a[j]>a[j+1]){t=a[j]; a[j]=a[j+1]; a[j+1]=t;}}}
for(i=0;i<n/2;i++)printf("%4d",a[i]);
for(i=n-1;i>=n/2;i--)printf("%4d",a[i]);
getch();
}
```

```
Enter 8 elements 3 -3 9 4 -8 1 6 5
  -8  -3   1   3   9   6   5   4
```



Inserting a new element into array at specified position [push / right shifting of array elements]:

```
#include<stdio.h> #include<conio.h>
void main()
{
int a[100],n,i,ele,pos; clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter new element and position ");
scanf("%d%d",&ele,&pos);
if(pos>n+1 || pos<1)
printf("Position 1 to %d only",n+1);
else{
for(i=n;i>=pos;i--)a[i]=a[i-1];/*Right shifting*/
a[i]=ele; puts("Elements ");
for(i=0;i<=n;i++)printf("%4d",a[i]);
}
getch();
}
```

```
Enter array size 1-100 3
Enter 3 elements 1 2 3
Enter new element and position 4 4
Elements
   1   2   3   4_
```

TC

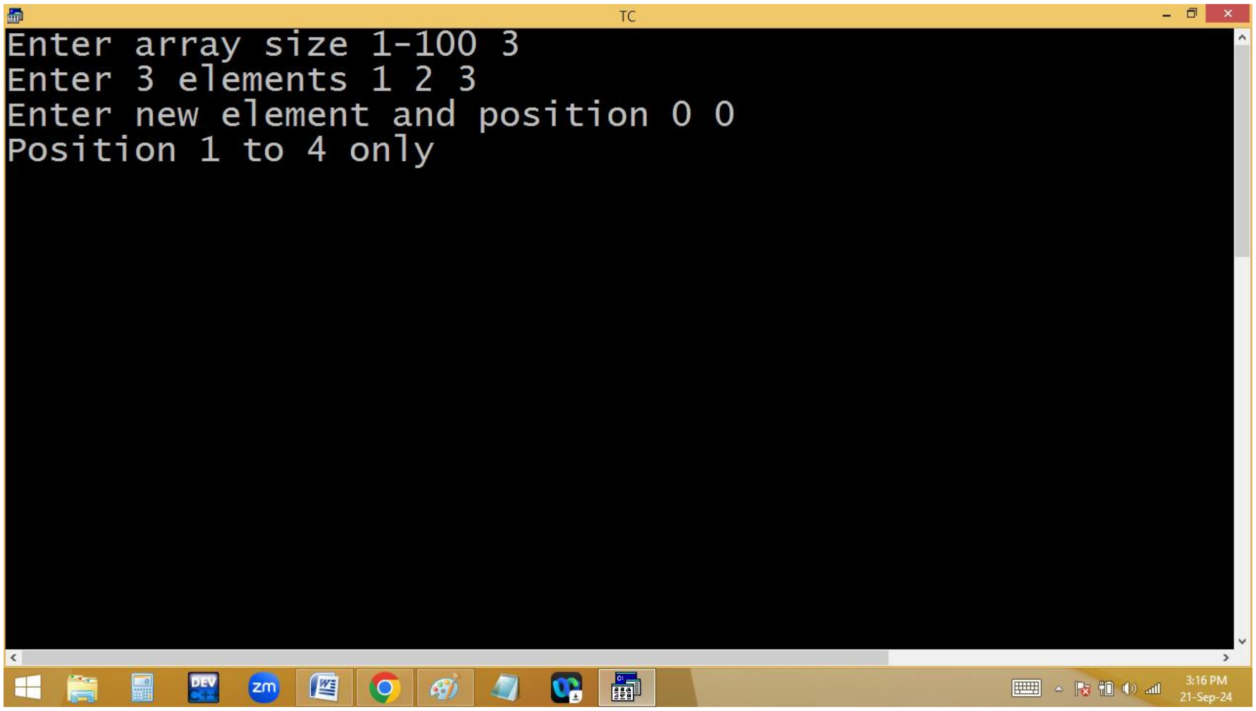
```
Enter array size 1-100 4
Enter 4 elements 1 2 4 5
Enter new element and position 3 3
Elements
   1   2   3   4   5
```

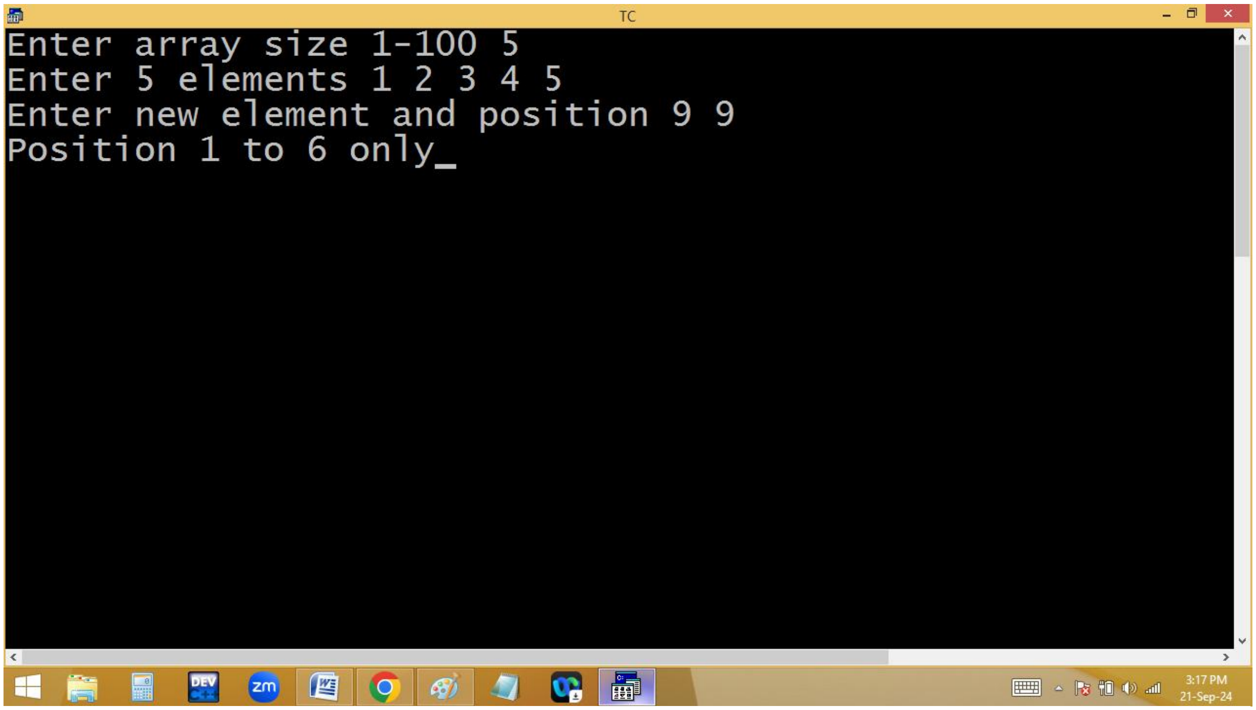
3:16 PM
21-Sep-24

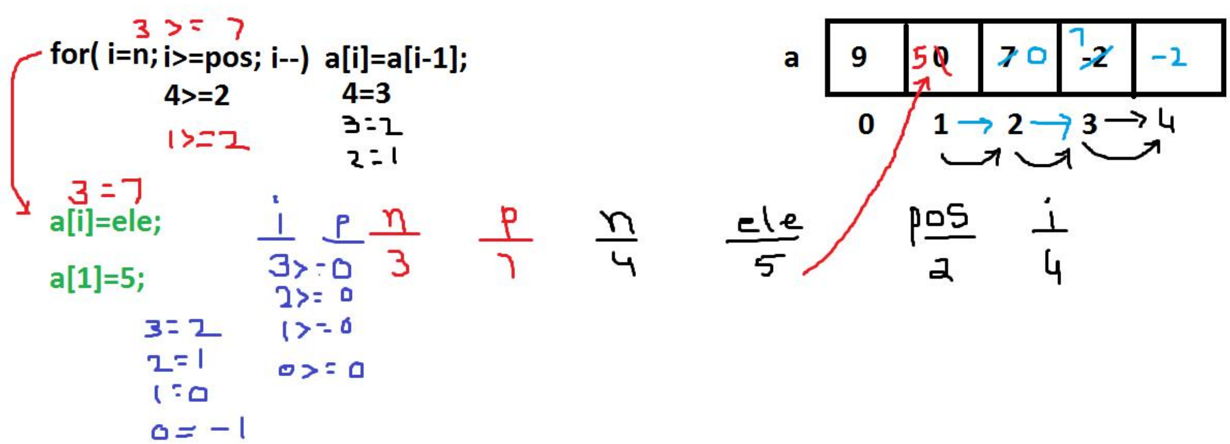
TC

```
Enter array size 1-100 3
Enter 3 elements 1 2 3
Enter new element and position 0 1
Elements
   0   1   2   3_
```

3:16 PM
21-Sep-24


TC
Enter array size 1-100 3
Enter 3 elements 1 2 3
Enter new element and position 0 0
Position 1 to 4 only
3:16 PM
21-Sep-24



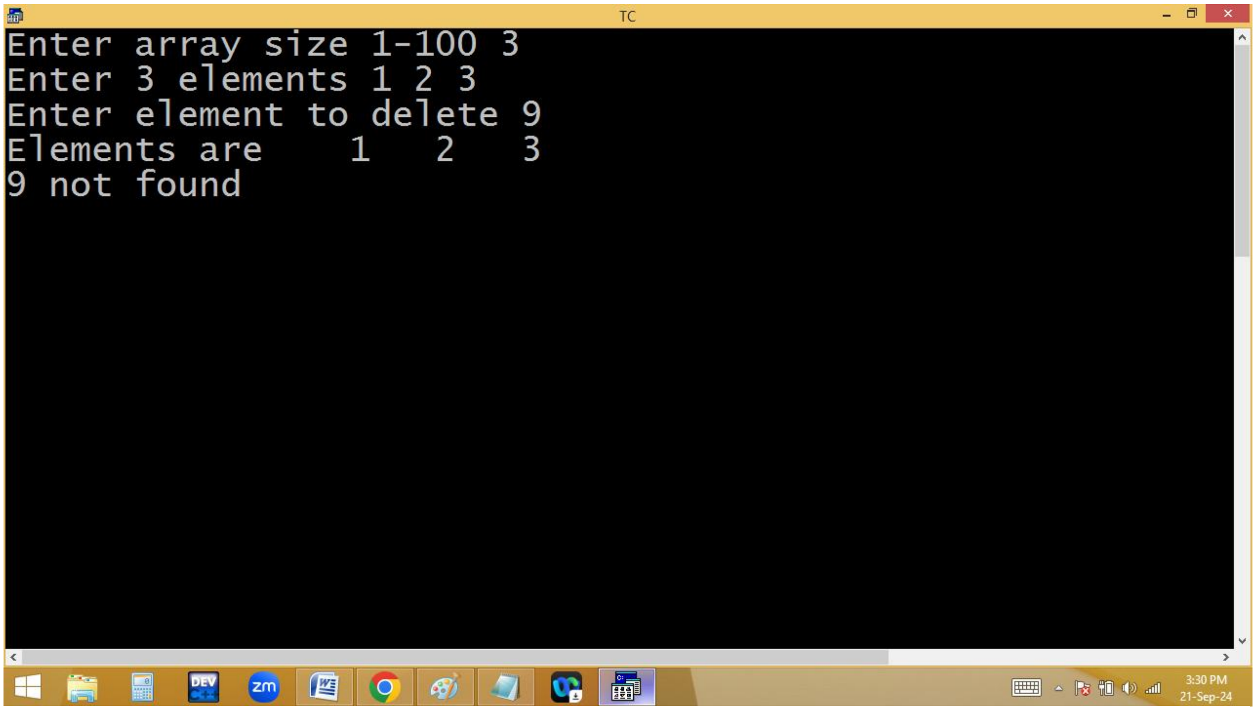


Deleting a particular element from array:

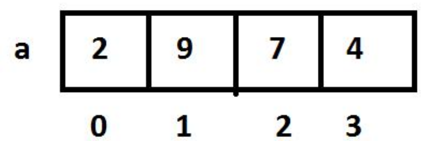
Skipping element:

```
File    Edit    Run    Compile    Project    Options    Debug
     Line 11    Col 11   Insert Indent Tab Fill Unindent
#include<stdio.h> #include<conio.h>
void main()
{
int a[100],n,i,ele,f=0; clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter element to delete ");
scanf("%d",&ele);
printf("Elements are ");
for(i=0;i<n;i++)if(a[i]!=ele)printf("%4d",a[i]);
else  f=1;
if(f==0)printf("\n%d not found", ele);
getch();
}
```

```
Enter array size 1-100 9
Enter 9 elements 1 2 3 1 4 5 1 6 7
Enter element to delete 1
Elements are    2   3   4   5   6   7_
```



for(i=0; i<4;i++)
if(a[i]!=ele)p(a[i]); 2 7 4

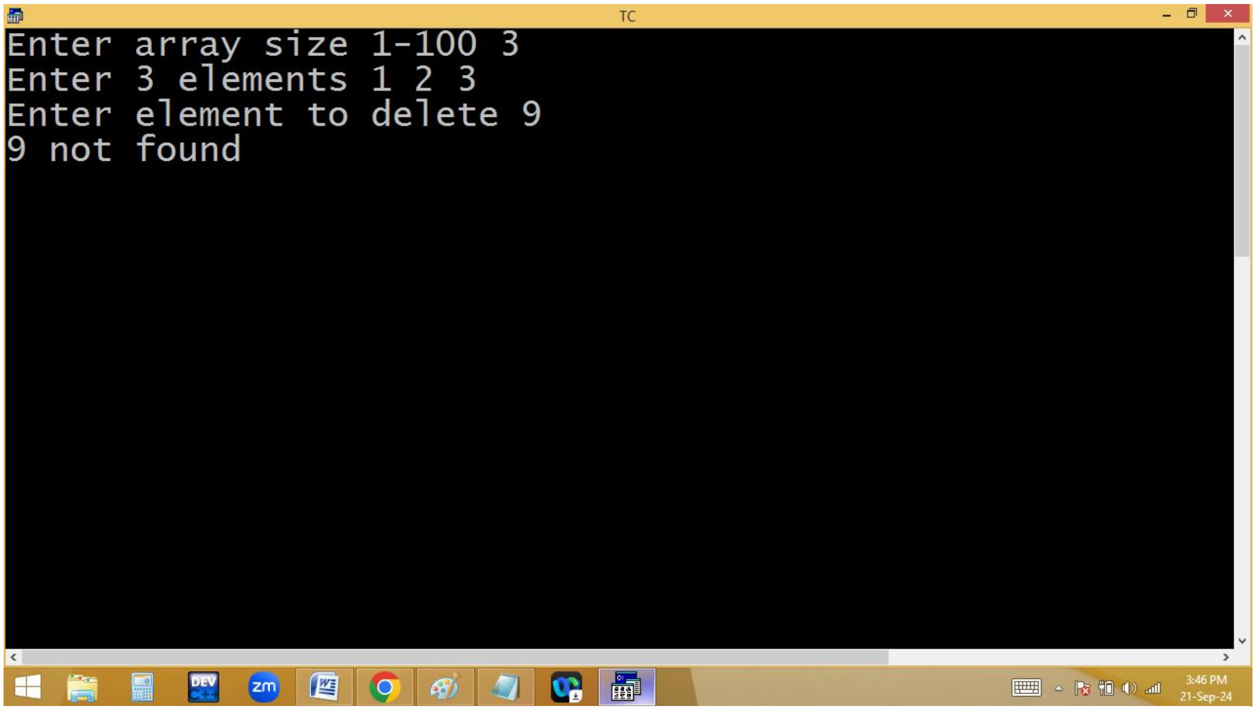


i	n	ele
0	4	9
1		
2		
3		

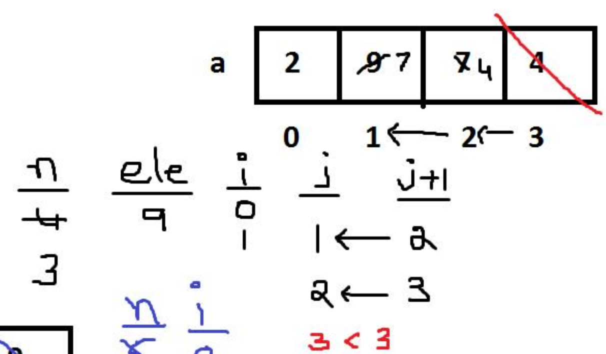
Permanent deletion:

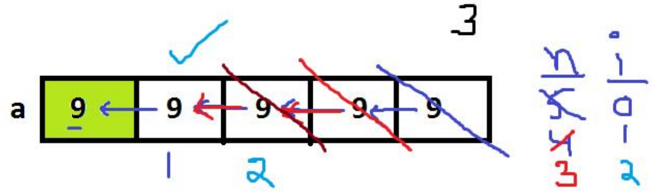
```c
#include<stdio.h> #include<conio.h>
void main()
{
int a[100],n,i,ele,f=0,j; clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter element to delete ");scanf("%d",&ele);
for(i=0;i<n;i++)
{
if(a[i]==ele){
for(n--,f=1,j=i;j<n;j++)a[j]=a[j+1];i--;}}
if(f==0)printf("%d not found", ele);
else {printf("Elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
}
getch();
}
```

```
Enter array size 1-100 5
Enter 5 elements 9 9 9 9 9
Enter element to delete 9
Elements are _
```




```
for(i=0;i<n;i++)
{
if(a[i]==ele)
{
for(n--,f=1 j=i ; j<n;j++) a[j]=a[j+1]; i--;
 }
 }
```





Home work:

1. Deleting duplicate elements from array
2. Finding 2nd max and min elements of array.
3. Finding nth max and nth min elements.